MATH 330—Proof, Set Theory, and Logic
Syllabus—Fall, 2026

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Class: MWF 10:10 AM–11:00 AM
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Location: Witmer 211
Office Hours: MW 11:00 AM–12:00 PM

Document Information: This syllabus describes the requirements and procedures for MATH 330—Proof, Set Theory, and Logic. You are responsible for knowing this material, so please read it carefully. Substantive changes will be announced via Blackboard announcement. You will be responsible for any changes. Your continued enrollment in this course is your implicit agreement to abide by the requirements of this class.

Prerequisites: MATH 166 or consent of instructor.

Textbook: *Elementary Set Theory, Fall 2026 Updated Version* by Richard Millspaugh, revised by Bryce Christopherson. A PDF copy of the book is available free of charge. This can be found under the "Textbook" section of the course Blackboard page.

Course Description: In this class, we will cover the following: Methods of proof, axioms and operations on sets, mathematical logic, relations and functions, development of the natural and real number systems, including field axioms and the completeness axiom for the real numbers.

Course Materials: The materials for this course, both required and optional, are as follows:

Textbook (Required): *Elementary Set Theory, Fall 2026 Updated Version* by Richard Millspaugh, revised by Bryce Christopherson. A PDF copy of the text can be found, free of cost, in the "Textbook" section of the course Blackboard page.

Calculator (optional): A scientific non-graphing calculator or computer equivalent is permitted, though not required, for this course.

Blackboard: All materials for this course will be posted to the class Blackboard page. Any updates or changes pertaining to the course will be posted as an announcement in Blackboard. Adjust your notification settings so that you receive emails or texts when announcements are posted. You want to be sure that you do not miss updates on assignments or other important information.
To access Blackboard, please go to: http://blackboard.UND.edu and log in with your NDUS Identifier (Username and Password). If you do not know your NDUS Identifier or have forgotten your password, please visit your NDUS account webpage on the University Information Technologies (UIT) website. For technical assistance, please contact UND Technical Support at 701.777.2222, visit the UIT website for their hours, help documents, and other resources, or visit the Knowledge Base for additional supports and information about general tech requirements for students including information about devices, operating systems, software, internet connection, and major-specific tech requirements.

Grading Policy: Your percentage grade is determined by the components detailed in the table below. You can estimate your letter grade by using the following scale: 90–100% is an A, 80–89% is a B, 70–79% is a C, 60–69% is a D, and 0–59% is an F. Plus/Minus Grades will not be awarded.

Category	Points	Sections Covered
Exam I	70 pts (25%)	Ch. 1, Ch.2,
Exam II	70 pts (25%)	Ch. 3, Ch. 4, Ch. 5
Final Exam	70 pts (25%)	Ch. 6, Ch. 7, Ch. 8
Homework	70 pts (25%)	
Total	280 pts (100%)	

Incompletes: It is expected that students will complete all requirements for a course during the time frame of the course. For reasons beyond a student's control, and upon request by the student or on behalf of the student, an incomplete grade may be assigned by the instructor when there is reasonable certainty the student will successfully complete the course without retaking it. The mark "I," Incomplete, will be assigned only to the student who has been in attendance and has done satisfactory work up to a time within four weeks of the close of the semester, including the examination period, and whose work is incomplete for reasons satisfactory to his or her instructor. More information regarding UND's Incomplete policy can be found on the UND "Grading System" webpage.

Exams: All exams except the final exam will be administered during the regularly scheduled class time. On examinations, you will be given a number of problems (generally, similar to those assigned in the homework) to complete during a fixed time period. To receive full credit on your submission, you are **required** to show **all** of your work in a **neat**, **coherent**, and **mathematically correct** fashion. A make-up exam will only be administered if there is documentation from a proper authority, such as a note from a physician in the case of illness. If you know that you will be unable to complete an exam in the allotted time, please talk to your instructor at least one week before the date the exam will be administered. If you have any questions about the scheduling of the exams, please discuss this with your instructor.

Make-up Exams: If you will not be able to take an exam on the scheduled date and time you must make other arrangements with the instructor well in advance. In cases where proper notification has not been given, or if the reason for missing the exam is not valid, the make-up exam may be penalized or may not be given.

Homework: Seven written homework assignments (equally weighted) will be given during the semester. Homework is intended as mathematical practice, not as a test of punctuality. Each assignment will have a suggested completion date, but late work will not be penalized. Complete solutions will be posted on an announced date regardless of who has submitted.

Each assignment will produce **two separate submissions** and **two separate grades** (each worth half of the final score): an **initial check** and a **proof audit**. Each initial check and its corresponding proof audit must be submitted separately. Proof audits will be assigned and collected throughout the semester rather than combined into a single end-of-semester submission.

All submitted work must be neat, coherent, legible, and sufficiently detailed to communicate your reasoning. The initial check is evaluated for good-faith effort and completeness; the proof audit is evaluated for mathematical correctness and the quality of your analysis and explanation.

Suggested dates for both components will appear in the course calendar and on Blackboard. All times are listed in the Central Time Zone. Although late work will not be penalized, all components of every assignment must be completed during the official course dates.

Initial Check: To receive points for the **initial check**, you must submit either a typed copy or a scanned copy of your handwritten first attempt through the class Blackboard page. You must complete this attempt without consulting the posted solutions, even if the solutions have already been released. Initial checks will be graded on a rolling basis. Full credit requires what can reasonably be considered a good-faith effort to complete every problem; partial credit will be awarded according to how much of the assignment meets this standard.

Proof Audit: After the suggested completion date, a solution key will be posted on the class Blackboard page. You should use this key to evaluate the correctness of your initial work and determine what needs to be repaired.

To receive points for the **proof audit**, you must separately audit one incorrect or incomplete proof from your initial attempt. Your audit should explain the intended strategy, compare your attempt with the correct reasoning, identify and explain a substantive gap or difficult step, repair that issue, and present a corrected argument.

If every proof in your initial submission was correct, you should instead select the proof you found most difficult. Your audit should identify its strategy and essential logical steps, explain why those steps are justified, and briefly reflect on the motivation for your reasoning and the proof techniques you employed.

Copying or paraphrasing the posted solution does not constitute an audit.

Extra Credit: Opportunities for earning extra credit may occur throughout the semester at the discretion of the instructor. Extra credit problems may be given in class or occasionally may be offered on exams. Points earned from extra credit may help borderline grades, but do not expect extra credit to significantly change your final score.

Attendance: Attending regularly scheduled lecture sessions and/or viewing lecture videos is not mandatory, but I believe it is very important. If you frequently do not attend/view the lectures without a good reason and begin to struggle in the course, I will have very little sympathy for you.

Getting Help: One of the most important skills a student can develop is the ability to recognize quickly when they are struggling with some concept in a class. Once you recognize you need help, first check through your materials (such as your textbook or class notes, for example) to see if the answer is there. If you cannot alleviate whatever difficulties you are having yourself, *please seek help.* There are lots of resources available to you. It is important that you try a variety of different services. If you do not receive the help you need at that time, please try it again or try a different service. Don't be shy—make sure you get the help you need!

Study Groups: Meet with some of your classmates and help each other with questions. Learning from your peers is one of the best ways to learn.

Office Hours: My office hours (subject to change as noted above) are listed at the top of this document. Please seek me outside of class if you have any questions, concerns, or would like any additional help with the course material. If these times are not convenient, we should be able to arrange an alternative time. I am here to help you!

UND Services: UND cares about your success as a student and provides a number of services, such as The UND Writing Center, Tutoring and Learning Services, Testing Services, and more. For additional information, visit the UND "Student Resources" webpage.

Academic Integrity: Academic integrity is a serious matter, and any deviations from appropriate behavior will be dealt with strongly. As a scholastic matter, the professor has the discretion to determine appropriate penalties for the student's workload or grade, but the situation may be resolved without involving many individuals. An alternative is to treat the situation as a disciplinary matter, which can result in suspension from the University, or have lesser penalties. Be aware that I view this as a very serious matter and will have little tolerance and/or sympathy for questionable practices. A student who attempts to obtain credit for work that is not their own (whether that be on a paper, quiz, homework assignment, exam, etc.) will likely receive a failing grade for that item of work, and at the professor's discretion, may also receive a failing grade in the course. For more information read the Code of Student Life.

Disability Statement: The University of North Dakota is committed to providing equal access to students with documented disabilities. To ensure access to this class and your program, please contact Student Disability Resources (SDR) to engage in a confidential discussion about accommodations. Accommodations are not provided retroactively. Students are encouraged to register with SDR at the start of their program. More information can be obtained by email (UND.sdr@UND.edu) or by phone at 701-777-2100.

Course Evaluation: Near the end of the semester, you will be asked to complete an online course evaluation form (SELFI). Your feedback on the course is extremely valuable to me. I read my students' comments carefully and use them to improve the course the next time I teach it. When the time comes, please let me know which aspects of the course helped you learn—and which aspects might be modified to help future students learn more effectively. Please note that the course evaluations are anonymous and that I won't see the results until after the grades for the course are submitted, allowing you to provide honest and constructive feedback. Throughout the semester if you have concerns or feedback, please reach out to schedule a time to discuss.

UND Syllabi Statements: Additional syllabi statements are provided by the Office of the Provost.

Standards for Written Mathematical Work The rubric below describes the general standards expected of written proofs and other mathematical arguments in this course. It is intended as a guideline rather than as a mechanical point-allocation formula. In particular, the **initial check** portion of a homework assignment is evaluated according to the good-faith completion standard described above, not according to this rubric. These standards are most directly relevant to exams, corrected proofs submitted as part of proof audits, and any other work evaluated for mathematical quality. Proof audits are also evaluated according to the comparison and reflection requirements given in the Homework section.

You should strive to produce arguments that are clear and well organized, use mathematical notation and terminology precisely, justify each substantive step, apply definitions and theorems correctly, and reach a valid conclusion. Practicing these habits on homework will make it more likely that your written work will meet the standards expected on exams. Please consult this rubric when preparing solutions, reviewing feedback, or asking questions about the evaluation of your work.

Characteristics of the Written Work	**Qualitative Description**	**Typical Scoring %**
(i) The work is clear, well organized, complete, and easy to follow. The overall proof strategy is apparent. (ii) Variables are properly introduced, and mathematical terminology, notation, and quantifiers are used precisely. (iii) Definitions and theorems are applied appropriately, and their hypotheses are verified whenever this is not immediate. (iv) Every substantive step is justified, all relevant cases are addressed, and the conclusion follows logically. The work may contain a minor typographical or computational error that does not affect the essential reasoning.	Exceptional	90–100
(i) The work is generally clear, organized, and essentially complete, although a few steps may be presented too briefly. (ii) Mathematical terminology and notation are usually used correctly, with only minor lapses or ambiguities. (iii) The proof uses an appropriate strategy and relevant definitions or theorems, although one hypothesis or transition may not be fully explained. (iv) The argument is essentially correct and demonstrates strong understanding. Any gap or error can be repaired locally without changing the main strategy.	Good	75–89
(i) A relevant strategy can be identified, but the work may be incomplete, poorly organized, or difficult to follow in places. (ii) Mathematical terminology and notation are used inconsistently, or some variables, quantifiers, or assumptions are left unclear. (iii) The argument contains a substantive gap, omitted case, unjustified inference, or misapplication of a result that requires nontrivial repair. (iv) The work contains significant correct reasoning and demonstrates partial understanding, but it does not yet constitute a complete proof.	Fair	60–74
(i) The work contains some relevant ideas, but the intended argument is incomplete or difficult to reconstruct. (ii) Definitions, terminology, or notation are frequently used incorrectly, and important hypotheses may not be recognized or checked. (iii) The argument contains several major gaps, unsupported assertions, circular reasoning, or logically invalid steps. (iv) The work demonstrates limited understanding and cannot be corrected merely by repairing one isolated step.	Poor	40–59
(i) The work presents no coherent strategy relevant to the problem or makes little meaningful progress. (ii) There is little or no correct use of the relevant definitions, terminology, or notation. (iii) The response gives an answer without justification, merely restates the problem, or consists primarily of irrelevant or contradictory work. (iv) The work demonstrates little or no understanding of the mathematical ideas needed to solve the problem.	Very Poor	0–39

Course Schedule: (Chapters Covered: All)

Week/Day	Su.	Mo.	Tu.	We.	Th.	Fr.	Sa.
Week	23 Aug.	24 Aug.	25 Aug.	26 Aug.	27 Aug.	28 Aug.	29 Aug.
1				Intro, 1.1		1.2	
Week	30 Aug.	31 Aug.	1 Sept.	2 Sept.	3 Sept.	4 Sept.	5 Sept.
2		1.2		1.3		1.3	
Week	6 Sept.	7 Sept.	8 Sept.	9 Sept.	10 Sept.	11 Sept.	12 Sept.
3		*Labor Day (No Class)*		1.4		2.1	
Week	13 Sept.	14 Sept.	15 Sept.	16 Sept.	17 Sept.	18 Sept.	19 Sept.
4		2.2		2.2		2.3 **HW 1**	
Week	20 Sept.	21 Sept.	22 Sept.	23 Sept.	24 Sept.	25 Sept.	26 Sept.
5		2.4		3.1		3.2	
Week	27 Sept.	28 Sept.	29 Sept.	30 Sept.	1 Oct.	2 Oct.	3 Oct.
6		Review **HW 2**		3.3		**Exam I (Ch. 1, 2)**	
Week	4 Oct.	5 Oct.	6 Oct.	7 Oct.	8 Oct.	9 Oct.	10 Oct.
7		3.3		3.4		4.1	
Week	11 Oct.	12 Oct.	13 Oct.	14 Oct.	15 Oct.	16 Oct.	17 Oct.
8		4.2		4.3		5.2 **HW 3**	

Week/Day	Su.	Mo.	Tu.	We.	Th.	Fr.	Sa.
Week	18 Oct.	19 Oct.	20 Oct.	21 Oct.	22 Oct.	23 Oct.	24 Oct.
9		5.2		5.3		5.4 **HW 4**	
Week	25 Oct.	26 Oct.	27 Oct.	28 Oct.	29 Oct.	30 Oct.	31 Oct.
10		6.1		6.2		6.3 **HW 5**	
Week	1 Nov.	2 Nov.	3 Nov.	4 Nov.	5 Nov.	6 Nov.	7 Nov.
11		Review		6.3		**Exam II** **(Ch. 3, 4, 5)**	
Week	8 Nov.	9 Nov.	10 Nov.	11 Nov.	12 Nov.	13 Nov.	14 Nov.
12		Flex Day		*Veteran's Day* *(No Classes)*		7.1	
Week	15 Nov.	16 Nov.	17 Nov.	18 Nov.	19 Nov.	20 Nov.	21 Nov.
13		7.1		7.2		7.3 **HW 6**	
Week	22 Nov.	23 Nov.	24 Nov.	25 Nov.	26 Nov.	27 Nov.	28 Nov.
14		Catch-Up Day		*Thanksgiving Break* *(No Classes)*	*Thanksgiving Break* *(No Classes)*	*Thanksgiving Break* *(No Classes)*	
Week	29 Nov.	30 Nov.	1 Dec.	2 Dec.	3 Dec.	4 Dec.	5 Dec.
15		7.3		7.3		8.1	
Week	6 Dec.	7 Dec.	8 Dec.	9 Dec.	10 Dec.	11 Dec.	12 Dec.
16		8.1		Review **HW 7**	*Last Day* *of Classes*		
Week	13 Dec.	14 Dec.	15 Dec.	16 Dec.	17 Dec.	18 Dec.	19 Dec.
17				**Final Exam** **(Ch. 6, 7, 8)** **10:15–12:15** **Witmer 211**			